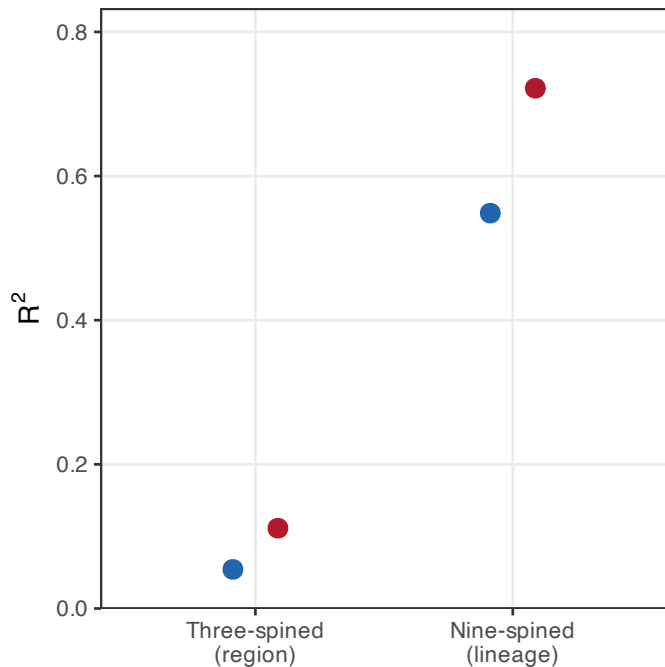
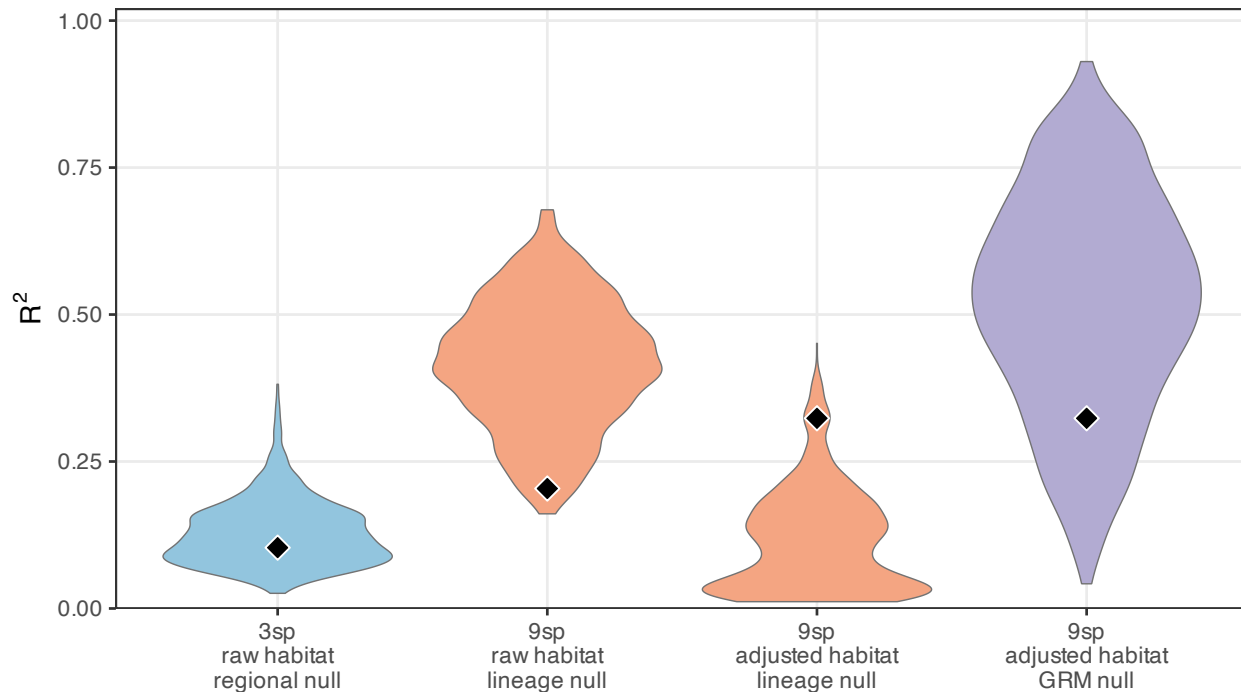


a**Alignment with null-defining groups****b****Alignment with five relationship-matrix axes**

Populations weighted equally



Individuals weighted equally



GRM-matched continuous null



Within-lineage label permutation



Within-region label permutation